

Introduction to Biophysics II: Quantum physics & biology

Lecture 11: Excitons in photosynthesis

Excitons

Reading:
Modern Optical Spectroscopy, chapters 8
Additional:
Excitons. By H. van Amerongen, L. Valkunas, R.
van Grondelle. World Scientific 2000

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Excitons

Förster theory approximations:

- Weak coupling, slow energy transfer (slower than vibrations)

What if the coupling is so strong and the transfer is so fast that it is impossible to say which pigment is excited?

Exciton: delocalized excitation, shared by two or more molecules

Excitonic interaction: an interaction that forces excitation to delocalize

Excitonic delocalization → absorption/emission spectra are not a simple superposition of the spectra of individual molecules

Frenkel exciton: an excited electron on the LUMO orbital and a respective hole in the HOMO orbitals are on the same molecule, as excitation migrates between molecules

Wannier exciton: electron and hole could be on different molecules

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Coupled dimer: ground state

Start from dimer, same as in Förster theory.

Ground state:

$$\Psi_A = \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a}$$

Excited state:

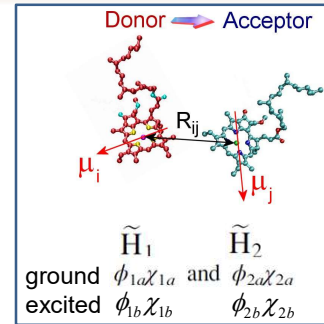
$$\begin{aligned}\Psi_B &= C_1 \overset{D^*A}{\psi_1} + C_2 \overset{DA^*}{\psi_2} \\ &= C_1 \overset{D^*}{\phi_{1b}}\overset{A}{\chi_{1b}}\overset{A}{\phi_{2a}}\overset{A}{\chi_{2a}} + C_2 \overset{D}{\phi_{1a}}\overset{A^*}{\chi_{1a}}\overset{A}{\phi_{2b}}\overset{A^*}{\chi_{2b}}\end{aligned}$$

Total Hamiltonian: $H = \tilde{H}_1 + \tilde{H}_2 + \tilde{H}'$
 $\tilde{H}'$ interaction

Ground state energy:

$$\begin{aligned}E_A &= \langle \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} | \tilde{H}_1 + \tilde{H}_2 + \tilde{H}' | \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} \rangle \\ &\quad \langle \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} | \tilde{H}_1 | \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} \rangle = \langle \phi_{1a}\chi_{1a} | \tilde{H}_1 | \phi_{1a}\chi_{1a} \rangle \langle \phi_{2a}\chi_{2a} | \phi_{2a}\chi_{2a} \rangle = E_{1a} \\ &\quad \langle \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} | \tilde{H}_2 | \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} \rangle = \langle \phi_{1a}\chi_{1a} | \phi_{1a}\chi_{1a} \rangle \langle \phi_{2a}\chi_{2a} | \tilde{H}_2 | \phi_{2a}\chi_{2a} \rangle = E_{2a}\end{aligned}$$

$$E_A = E_{1a} + E_{2a} + \langle \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} | \tilde{H}' | \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} \rangle$$



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Coupled dimer: excited state

Ground: $\Psi_A = \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a}$

Excited: $\Psi_B = C_1\psi_1 + C_2\psi_2$

$$= C_1\phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} + C_2\phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b}$$

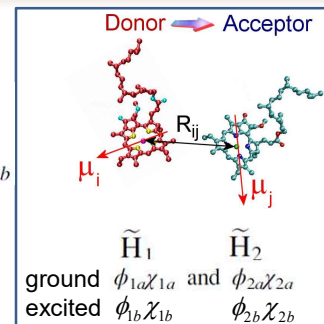
Total Hamiltonian: $H = \tilde{H}_1 + \tilde{H}_2 + \tilde{H}'$

Ground state energy:

$$E_A = E_{1a} + E_{2a} + \langle \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} | \tilde{H}' | \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} \rangle$$

Excited state energy:

$$E_B = \langle C_1\psi_1 + C_2\psi_2 | \tilde{H}_1 + \tilde{H}_2 + \tilde{H}' | C_1\psi_1 + C_2\psi_2 \rangle$$



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Coupled dimer: excited state

Excited state energy:

$$\begin{aligned}\Psi_B &= C_1\psi_1 + C_2\psi_2 \\ &= C_1\phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} + C_2\phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b} \\ E_B &= \langle C_1\psi_1 + C_2\psi_2 | \tilde{H}_1 + \tilde{H}_2 + \tilde{H}' | C_1\psi_1 + C_2\psi_2 \rangle \\ &\quad \langle C_1\psi_1 | \tilde{H}_1 | C_1\psi_1 \rangle = C_1^2 \langle \phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} | \tilde{H}_1 | \phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} \rangle = C_1^2 E_{1b} \\ &\quad \langle C_1\psi_1 | \tilde{H}_1 | C_2\psi_2 \rangle = C_1 C_2 \langle \phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} | \tilde{H}_1 | \phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b} \rangle = ? \\ &\quad \langle C_2\psi_2 | \tilde{H}_1 | C_2\psi_2 \rangle = C_2^2 \langle \phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b} | \tilde{H}_1 | \phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b} \rangle = C_2^2 E_{1a}\end{aligned}$$

Do the same for H_2 and combine:

$$E_B = C_1^2 E_1 + C_2^2 E_2 + \langle C_1\psi_1 + C_2\psi_2 | \tilde{H}' | C_1\psi_1 + C_2\psi_2 \rangle$$

$\uparrow$ $E_1 \equiv E_{1b} + E_{2a}$ $\uparrow$ $E_2 \equiv E_{2b} + E_{1a}$

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Coupled dimer: transition energies

Ground state energy:

$$E_A = E_{1a} + E_{2a} + \langle \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} | \tilde{H}' | \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} \rangle$$

Excited state energy:

$$E_B = C_1^2 E_1 + C_2^2 E_2 + \langle C_1\psi_1 + C_2\psi_2 | \tilde{H}' | C_1\psi_1 + C_2\psi_2 \rangle$$

$\uparrow$ $E_1 \equiv E_{1b} + E_{2a}$ $\uparrow$ $E_2 \equiv E_{2b} + E_{1a}$

A special case of no interaction:

What are transition energies for states 1 and 2? ($C_1=1$ or $C_2=1$)

Interaction:

$\langle \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} | \tilde{H}' | \phi_{1a}\chi_{1a}\phi_{2a}\chi_{2a} \rangle$ - **Displacement** of ground state energy shifts both transition energies equally. Often ignored since it is small and we are mostly interested in splitting between transition energy levels.

$\langle C_1\psi_1 + C_2\psi_2 | \tilde{H}' | C_1\psi_1 + C_2\psi_2 \rangle$ - Shifts transition energies in respect to each other, can be significant!

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Dimer: excited state SE

Solve Schrödinger equation for new stationary states:

→ Seek coefficients C_1 and C_2 so that we get new stationary states of perturbed Hamiltonian

Excited state energy of new stationary state

$$(\tilde{H}_1 + \tilde{H}_2 + \tilde{H}') (C_1 \psi_1 + C_2 \psi_2) = E_B (C_1 \psi_1 + C_2 \psi_2)$$

Multiply with ψ_1^* from left and integrate over all space:

$$C_1 [\langle \psi_1 | \tilde{H}_1 | \psi_1 \rangle + \langle \psi_1 | \tilde{H}' | \psi_1 \rangle] + C_2 \langle \psi_1 | \tilde{H}' | \psi_2 \rangle = E_B [C_1 \langle \psi_1 | \psi_1 \rangle + C_2 \langle \psi_1 | \psi_2 \rangle] \\ = E_B C_1$$

And we get:

$$C_1 [\langle \psi_1 | \tilde{H}_1 | \psi_1 \rangle + \langle \psi_1 | \tilde{H}' | \psi_1 \rangle - E_B] + C_2 \langle \psi_1 | \tilde{H}' | \psi_2 \rangle = 0$$

Similarly, multiply with ψ_2^* from the left and integrate over all space:

$$C_1 \langle \psi_2 | \tilde{H}' | \psi_1 \rangle + C_2 [\langle \psi_2 | \tilde{H}_2 | \psi_2 \rangle + \langle \psi_2 | \tilde{H}' | \psi_2 \rangle - E_B] = 0$$

Got system of 2 equations

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Dimer: excited state SE

Solve Schrödinger equation for new stationary states:

$$\begin{cases} C_1 [\langle \psi_1 | \tilde{H}_1 | \psi_1 \rangle + \langle \psi_1 | \tilde{H}' | \psi_1 \rangle - E_B] + C_2 \langle \psi_1 | \tilde{H}' | \psi_2 \rangle = 0 \\ C_1 \langle \psi_2 | \tilde{H}' | \psi_1 \rangle + C_2 [\langle \psi_2 | \tilde{H}_2 | \psi_2 \rangle + \langle \psi_2 | \tilde{H}' | \psi_2 \rangle - E_B] = 0 \end{cases}$$

Rewrite:

$$\begin{cases} C_1 (H_{11} - E_B) + C_2 H_{12} = 0 \\ C_1 H_{21} + C_2 (H_{22} - E_B) = 0 \end{cases}$$

$$H_{11} \equiv \langle \psi_1 | \tilde{H}_1 + \tilde{H}' | \psi_1 \rangle = E_1 + \langle \psi_1 | \tilde{H}' | \psi_1 \rangle$$

$$H_{22} \equiv \langle \psi_2 | \tilde{H}_2 + \tilde{H}' | \psi_2 \rangle = E_2 + \langle \psi_2 | \tilde{H}' | \psi_2 \rangle$$

$$H_{21} \equiv \langle \psi_2 | \tilde{H}' | \psi_1 \rangle, \quad H_{12} \equiv \langle \psi_1 | \tilde{H}' | \psi_2 \rangle$$

Note: Hermitian operator → $H_{12} = H_{21}^*$.

$$\begin{bmatrix} \langle \psi_1 | \tilde{H}' | \psi_1 \rangle \\ \langle \psi_2 | \tilde{H}' | \psi_2 \rangle \end{bmatrix}$$

← Show how interactions affect the excited state energies of molecules 1 and 2, respectively. Often small compared to E_1 and E_2

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Dimer: excited state SE

Solve Schrödinger equation for new stationary states:

$$\begin{cases} C_1(H_{11} - E_B) + C_2H_{12} = 0 \\ C_1H_{21} + C_2(H_{22} - E_B) = 0 \end{cases} \quad \begin{aligned} \Psi_B &= C_1\psi_1 + C_2\psi_2 \\ &= C_1\phi_{1b}\chi_{1b}\phi_{2a}\chi_{2a} + C_2\phi_{1a}\chi_{1a}\phi_{2b}\chi_{2b} \end{aligned}$$

$$\text{Also: } C_1^2 + C_2^2 = 1$$

Math: non-trivial (i.e., C_1 and C_2 are not zeroes) only exists if determinant:

$$\begin{vmatrix} (H_{11} - E_B) & H_{12} \\ H_{21} & (H_{22} - E_B) \end{vmatrix} = 0$$

(See Box 8.1 in a textbook for more on that math)

$$\text{Solving for } E_B: (H_{11} - E_B)(H_{22} - E_B) - H_{12}^2 = 0$$

$$E_B^2 + (-H_{11} - H_{22})E_B + H_{11}H_{22} - H_{12}^2 = 0$$

$$2 \text{ roots: } E_{B+} = E_0 + \frac{1}{2}\sqrt{\delta^2 + 4(H_{12})^2}, \quad E_0 = (H_{11} + H_{22})/2$$

$$E_{B-} = E_0 - \frac{1}{2}\sqrt{\delta^2 + 4(H_{12})^2}, \quad \delta = (H_{11} - H_{22})$$

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Dimer: excited state SE

Or in matrix form:

$$\text{Our wavefunction basis forms a vector: } \psi_{\text{basis}} = (\psi_1 \quad \psi_2)$$

$$\text{Our mixed wavefunction: } \psi_B = C_1\psi_1 + C_2\psi_2 \rightarrow \psi_B = \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\text{Hamiltonian: } \mathbf{H} = \begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix}$$

$$\text{Schrödinger: } \mathbf{H}\psi_B = E_B\psi_B$$

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E_B \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$

$$\begin{pmatrix} H_{11} - E_B & H_{12} \\ H_{21} & H_{22} - E_B \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = 0$$

This is the same matrix as we had...

A nontrivial solution requires the determinant to be zero. That defines E_B

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Dimer: excited state SE

Solve Schrödinger equation for new stationary states:

$$\begin{vmatrix} (H_{11} - E_B) & H_{21} \\ H_{21} & (H_{22} - E_B) \end{vmatrix} = 0$$



$$\begin{aligned} E_{B\pm} &= E_0 \pm H_{12} \sqrt{\frac{\delta^2}{4H_{12}^2} + 1} \\ E_0 &= (H_{11} + H_{22})/2 \\ \delta &= (H_{11} - H_{22}) \end{aligned}$$

What about C_1 and C_2 ?

Solve for each E_B

$$\begin{pmatrix} H_{11} & H_{12} \\ H_{21} & H_{22} \end{pmatrix} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = E_B \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}$$



One more equation:

$$C_1^2 + C_2^2 = 1$$

$$\Psi_{B+} = \sqrt{\frac{(1+s)}{2}} \psi_1 + \sqrt{\frac{(1-s)}{2}} \psi_2$$

$$\Psi_{B-} = \sqrt{\frac{(1-s)}{2}} \psi_1 - \sqrt{\frac{(1+s)}{2}} \psi_2$$

$$s = \delta / \sqrt{\delta^2 + 4(H_{12})^2},$$

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Dimer: excited state SE

Analyze solutions

No interaction:

$$\Psi_{B+} = \sqrt{\frac{(1+s)}{2}} \psi_1 + \sqrt{\frac{(1-s)}{2}} \psi_2$$

$$\Psi_{B-} = \sqrt{\frac{(1-s)}{2}} \psi_1 - \sqrt{\frac{(1+s)}{2}} \psi_2$$

$$s = \delta / \sqrt{\delta^2 + 4(H_{12})^2}, \text{ =1}$$

$$E_{B\pm} = E_0 \pm H_{12} \sqrt{\frac{\delta^2}{4H_{12}^2} + 1}$$

$$E_0 = (H_{11} + H_{22})/2$$

$$\delta = (H_{11} - H_{22})$$

$$\Psi_{B+} = ? \quad = \psi_1$$

$$E_{B+} = ? \quad = H_{11}$$

$$\Psi_{B-} = ? \quad = \psi_2$$

$$E_{B-} = ? \quad = H_{22}$$

$$E_{B\pm} = E_0 \pm \sqrt{\frac{H_{12}^2 \delta^2}{4H_{12}^2} + H_{12}^2} = E_0 \pm \frac{\delta}{2}$$

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Dimer: excited state SE

Analyze solutions

Interaction, homodimer ($H_{11}=H_{22}$):

$$\Psi_{B+} = \sqrt{\frac{(1+s)}{2}} \psi_1 + \sqrt{\frac{(1-s)}{2}} \psi_2$$

$$\Psi_{B-} = \sqrt{\frac{(1-s)}{2}} \psi_1 - \sqrt{\frac{(1+s)}{2}} \psi_2$$

$$s = \delta / \sqrt{\delta^2 + 4(H_{12})^2}, = 0$$

$$E_{B\pm} = E_0 \pm H_{12} \sqrt{\frac{\delta^2}{4H_{12}^2} + 1}$$

$$E_0 = (H_{11} + H_{22})/2 = H_{11}$$

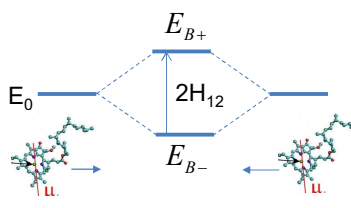
$$\delta = (H_{11} - H_{22}) = 0$$

$$\psi_{B+} = ? = \sqrt{1/2}(\psi_1 + \psi_2)$$

$$E_{B+} = ? = E_0 + H_{12}$$

$$\psi_{B-} = ? = \sqrt{1/2}(\psi_1 - \psi_2)$$

$$E_{B-} = ? = E_0 - H_{12}$$



Excited states split when molecules get close!

$E_{B\pm} = E_0 \pm H_{12}$ Note: + or - states can change order depending on sign of interaction

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Dimer: excited state SE

Analyze solutions

Interaction, $\delta \gg H_{12}$

$$\Psi_{B+} = \sqrt{\frac{(1+s)}{2}} \psi_1 + \sqrt{\frac{(1-s)}{2}} \psi_2$$

$$\Psi_{B-} = \sqrt{\frac{(1-s)}{2}} \psi_1 - \sqrt{\frac{(1+s)}{2}} \psi_2$$

$$s = \delta / \sqrt{\delta^2 + 4(H_{12})^2}, \approx 1$$

$$E_{B\pm} = E_0 \pm H_{12} \sqrt{\frac{\delta^2}{4H_{12}^2} + 1}$$

$$E_0 = (H_{11} + H_{22})/2$$

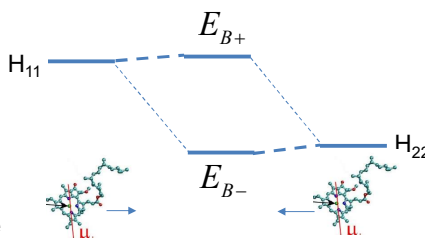
$$\delta = (H_{11} - H_{22})$$

$$\psi_{B+} = ? \approx \psi_1$$

$$E_{B+} = ? \approx H_{11}$$

$$\psi_{B-} = ? \approx \psi_2$$

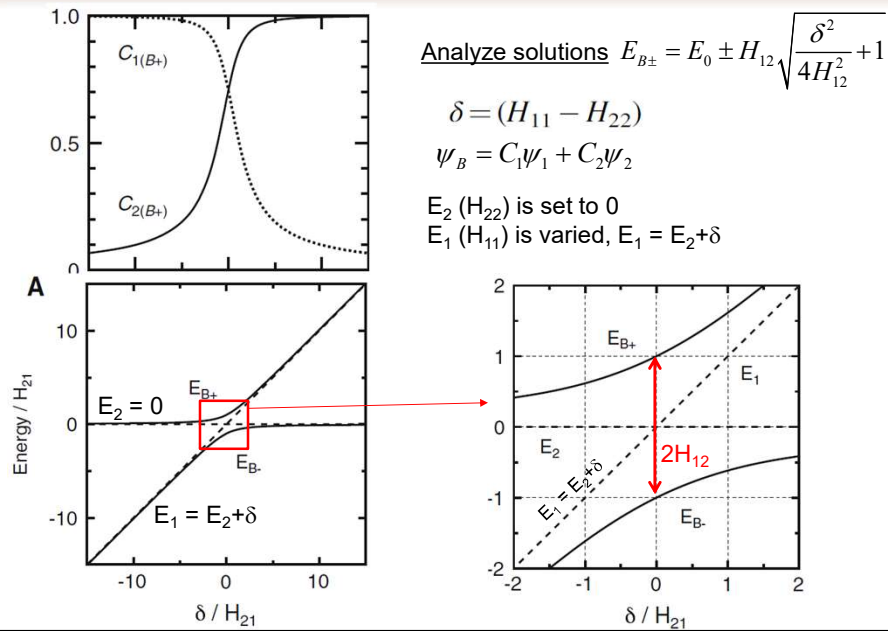
$$E_{B-} = ? \approx H_{22}$$



Excited states "repulse" a little
Are mostly localized on one pigment

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Dimer: excited state SE



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Dimer: excited state SE

Analyze solutions

$\Psi_{B+} = \sqrt{\frac{(1+s)}{2}} \psi_1 + \sqrt{\frac{(1-s)}{2}} \psi_2$ $\Psi_{B-} = \sqrt{\frac{(1-s)}{2}} \psi_1 - \sqrt{\frac{(1+s)}{2}} \psi_2$ $s = \delta / \sqrt{\delta^2 + 4(H_{12})^2},$	$E_{B\pm} = E_0 \pm H_{12} \sqrt{\frac{\delta^2}{4H_{12}^2} + 1}$ $E_0 = (H_{11} + H_{22})/2$ $\delta = (H_{11} - H_{22})$
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Notes:

- Ψ_{B+} and Ψ_{B-} states are **stationary** states; they are **orthonormal**
- H' interaction causes oscillations between ψ_1 and ψ_2 , but not Ψ_{B+} and Ψ_{B-} :

$$\langle \Psi_{B+} | \tilde{H}' | \Psi_{B-} \rangle = \langle \Psi_{B-} | \tilde{H}' | \Psi_{B+} \rangle = 0$$
- States Ψ_{B+} and Ψ_{B-} will remain stationary as long as no other perturbation is present.

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Exciton absorption spectra

New excitonic delocalized states: $\Psi_{B\pm} = C_1\psi_1 + C_2\psi_2$

Need to derive transition dipole moments for each excitonic state

$$\begin{aligned}\mu_{BA\pm} &= \langle \Psi_{B\pm} | \tilde{\mu}_1 + \tilde{\mu}_2 | \Psi_A \rangle \\ &= \langle C_{1(B\pm)} \phi_{1b} \phi_{2a} + C_{2(B\pm)} \phi_{1a} \phi_{2b} | \tilde{\mu}_1 + \tilde{\mu}_2 | \phi_{1a} \phi_{2a} \rangle \\ &= C_{1(B\pm)} \langle \phi_{1b} | \tilde{\mu}_1 | \phi_{1a} \rangle + C_{2(B\pm)} \langle \phi_{2b} | \tilde{\mu}_2 | \phi_{2a} \rangle\end{aligned}$$

$$\mu_{BA\pm} = C_{1(B\pm)} \mu_1 + C_{2(B\pm)} \mu_2$$

Transition dipole moment is a vector sum of transition dipole moments scaled with coefficients C_k . Absorption is proportional square of the transition dipole moment:

$$D_{BA\pm} = (C_{1(B\pm)})^2 D_{ba(1)} + (C_{2(B\pm)})^2 D_{ba(2)} + 2C_{1(B\pm)} C_{2(B\pm)} \mu_{ba(1)} \cdot \mu_{ba(2)}$$

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Exciton absorption spectra

Example: homodimer

$$\mu_{BA\pm} = C_{1(B\pm)} \mu_1 + C_{2(B\pm)} \mu_2$$

$$\Psi_{B+} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \Psi_{B-} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\vec{\mu}_{BA\pm} = \frac{1}{\sqrt{2}} (\vec{\mu}_{BA(1)} \pm \vec{\mu}_{BA(2)})$$

Dipole strength of each transition:

$$D_{BA\pm} \equiv |\vec{\mu}_{BA\pm}|^2 = \frac{1}{2} \left(|\vec{\mu}_{BA(1)}|^2 + |\vec{\mu}_{BA(2)}|^2 \pm 2\vec{\mu}_{BA(1)} \cdot \vec{\mu}_{BA(2)} \right)$$

$$D_{BA} \equiv |\vec{\mu}_{BA(1)}|^2 = |\vec{\mu}_{BA(2)}|^2$$

$|\mu|^2$ of individual molecule

$$D_{BA\pm} = \frac{1}{2} (2D_{BA} \pm D_{BA} \cos \theta)$$

$$D_{BA\pm} = D_{BA} (1 \pm \cos \theta)$$

Angle between μ 's of individual molecules

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Exciton absorption spectra

Example: homodimer

$$\vec{\mu}_{BA\pm} = \frac{1}{\sqrt{2}} (\vec{\mu}_{BA(1)} \pm \vec{\mu}_{BA(2)})$$

$$D_{BA\pm} = D_{BA} (1 \pm \cos \theta)$$

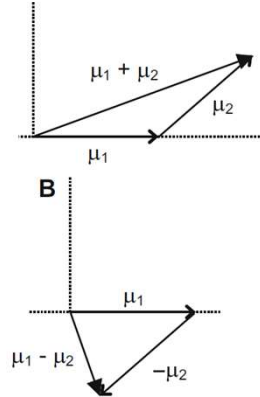
Angle:

$$\begin{aligned} \vec{\mu}_{BA+} \cdot \vec{\mu}_{BA-} &= \frac{1}{2} (\vec{\mu}_{BA(1)} + \vec{\mu}_{BA(2)}) \cdot (\vec{\mu}_{BA(1)} - \vec{\mu}_{BA(2)}) \\ &= \frac{1}{2} (|\vec{\mu}_{BA(1)}|^2 - |\vec{\mu}_{BA(2)}|^2) = ? \end{aligned}$$

Notes:

- Two excitonic transitions are always orthogonal
- Two bands have different absorption strengths (in general)
- Total transition strength for both bands is equal to that of two molecules (Transition dipole strength is redistributed between the two bands)
- Dipole-dipole interaction (H_{12}) also depends on that angle!

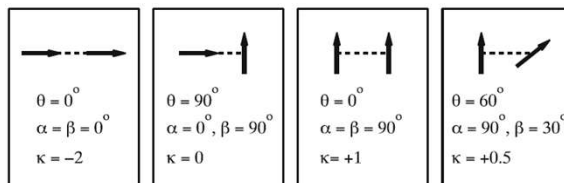
$$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$$



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Exciton absorption spectra

Example: homodimer



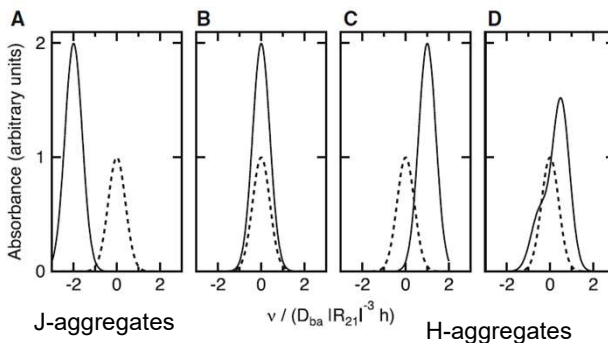
$$E_{B\pm} = E_0 \pm H_{12}$$

$$V_{12} = \frac{f^2}{n^2} \frac{\mu_2 \mu_1 \kappa}{R_{21}^3}$$

$$\kappa \equiv \cos \theta - 3 \cos \alpha \cos \beta$$

Dipole-dipole interaction

$$\vec{\mu}_{BA\pm} = \frac{1}{\sqrt{2}} (\vec{\mu}_{BA(1)} \pm \vec{\mu}_{BA(2)})$$



Solid: dimer

Dotted: monomer

Which is better for energy transfer, J- or H-aggregates?

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Multimers

Assume N strongly interacting molecules

Select basis – N excited state wavefunctions of *noninteracting* molecules $\phi_1, \phi_2, \dots, \phi_N$

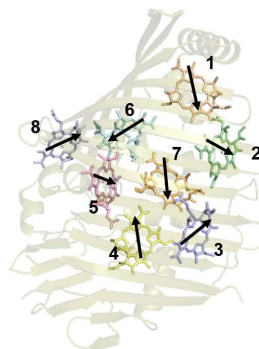
Represent *excitonic* excited state wavefunction as a superposition (projection to basis):

$$\psi_e = \begin{pmatrix} c_1 \\ c_2 \\ \vdots \\ c_N \end{pmatrix}$$

Hamiltonian:
$$\mathbf{H} = \begin{pmatrix} H_{11} & H_{12} & \cdots & H_{1N} \\ H_{21} & H_{22} & & H_{2N} \\ \vdots & & \ddots & \vdots \\ H_{N1} & H_{N2} & \cdots & H_{NN} \end{pmatrix}$$

Solve Schrödinger for stationary states: $\mathbf{H}\psi_e = E_e\psi_e$

Mathematical solution is equivalent to **diagonalization** of Hamiltonian matrix.



Fenna Matthews Olson complex: 8 BChl *b*

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Multimers

Solving $\mathbf{H}\psi_e = E_e\psi_e$

Hamiltonian:

$$\mathbf{H} = \begin{pmatrix} H_{11} & H_{12} & \cdots & H_{1N} \\ H_{21} & H_{22} & & H_{2N} \\ \vdots & & \ddots & \vdots \\ H_{N1} & H_{N2} & \cdots & H_{NN} \end{pmatrix}$$

Diagonalization:

Find matrix $\mathbf{C}$ so that:



$$\mathbf{C}^{-1}\mathbf{H}\mathbf{C} = \begin{pmatrix} E_1 & 0 & \cdots & 0 \\ 0 & E_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & E_N \end{pmatrix}$$

Energies $E_1 \dots E_N$ are eigenvalues of diagonalization

Columns of $\mathbf{C}$ are eigenvectors of diagonalization (expansion coefficients c_{ik})

Excitonic state i :

Energy:

$$E_i$$

Wavefunction:

$$\psi_i = \sum_{k=1}^N c_{ik} \phi_k$$

Transition dipole:

$$\vec{\mu}_i = \sum_{k=1}^N c_{ik} \vec{\mu}_{k,1 \text{ pigment}}$$

Python:

`numpy.linalg.eigh()`

Can diagonalize matrices $N \sim 10,000$

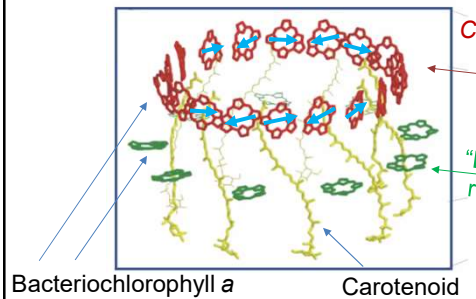
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Lecture 15

Example: LH2 complex

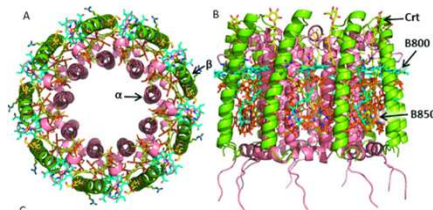
LH2 complex of photosynthetic purple sulfur bacteria

Photosynthetic antennae – e.g. LH1/RC, LH2



Bacteriochlorophyll a

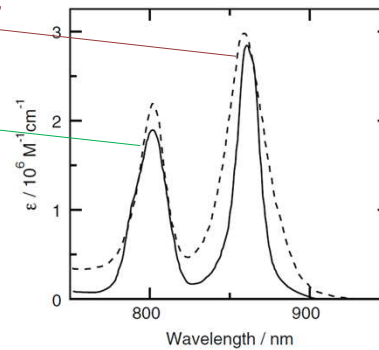
Carotenoid



Absorption

Solid: calculated

Dashed: measured



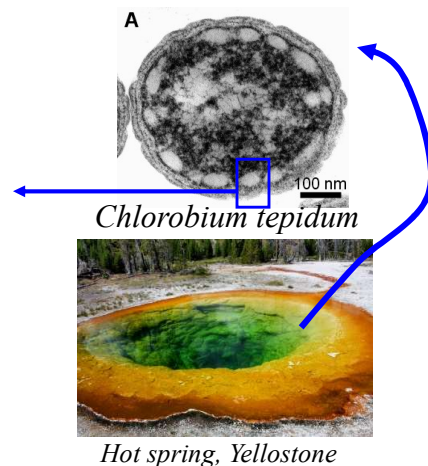
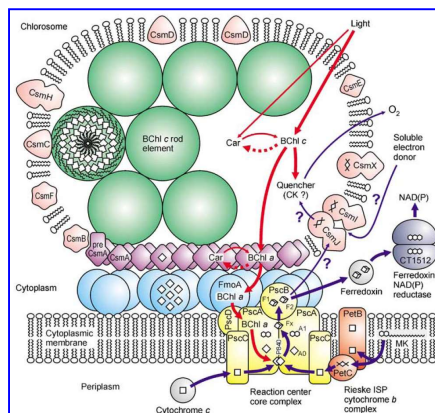
B850 absorption is red shifted due to excitonic coupling, and the lowest 3 exciton band have most of the transition dipole strength of 9x2 BChl a's, the rest bluer 15 bands are dark

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Lecture 09

Example: FMO complex

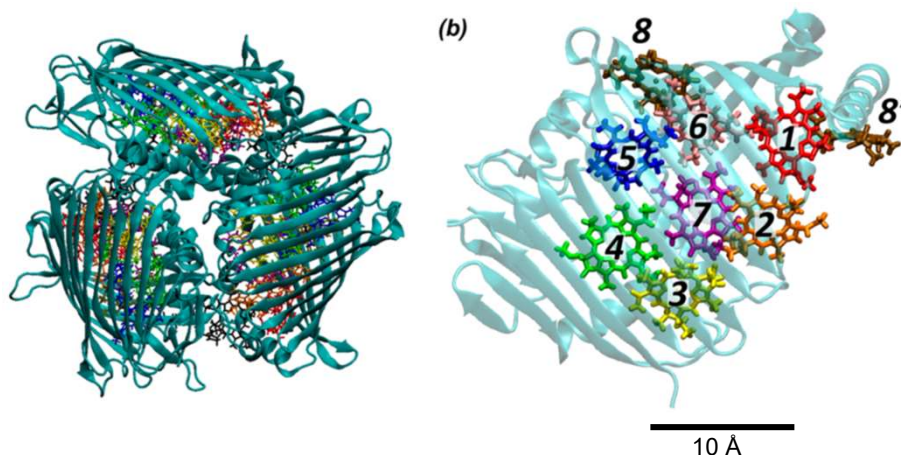
- Example: chlorosomes
- Common light harvesting antenna complex in green bacteria
- The most efficient antenna known: **200,000 (!)** bacteriochlorophylls



From Frigaard, N. et al. 2003 Photosyn. Res. 78: 93-117

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Example: FMO complex

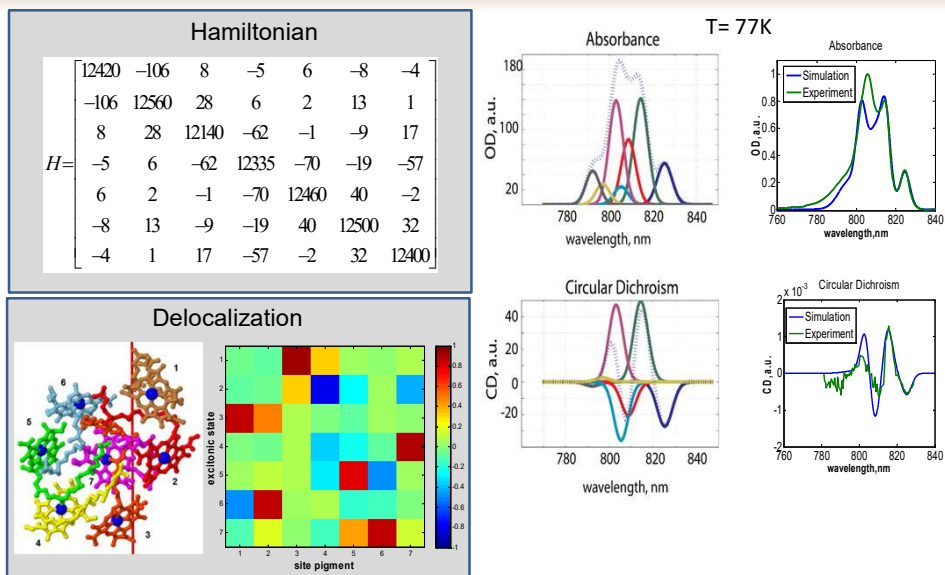


- First complex with x-ray structure (Fenna, Matthews 1975)
- Quantum Beats (Savikhin, Buck, Struve 1997)
- 2DES spectroscopy, coherences/vibrations (Brixner, ..., Fleming 2005)

(Structure from: Tronrud, Wen, Gay, & Blankenship, 2009)

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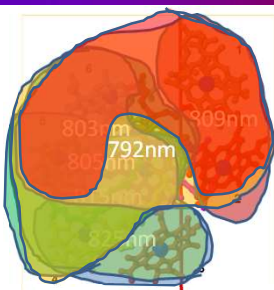
Example: FMO complex



Hamiltonian: Brixner et al. 2005, *Nature* **434**, 625-628

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The Hamiltonian



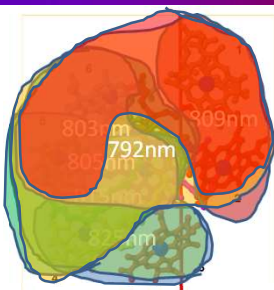
Singlet excited states are delocalized

Diagonal energies & interactions:
S. I. E. Vulto et al, Journal of Physical Chemistry B 102, 9577 (1998)

$E_{\text{site(pigment)}}$	E_i	c_{i1}	c_{i2}	c_{i3}	c_{i4}	c_{i5}	c_{i6}	c_{i7}
793.7(2)	790.272	-0.33	.77	.04	-.09	.29	.42	-.09
796.8(6)	791.708	.21	-.49	-.02	-.18	.53	.61	.13
800(5)	803.269	-.05	-.06	.04	-.09	-.56	.25	.78
804.5(7)	804.07	.05	.06	.13	-.38	.48	-.6	.49
806.5(1)	809.373	.91	.38	.08	-.02	-.1	.06	-.02
814.3(4)	815.534	.07	.03	-.36	.82	.27	-.09	.33
823.7(3)	825.615	-.05	-.07	.92	.37	.06	.02	.01

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The Hamiltonian



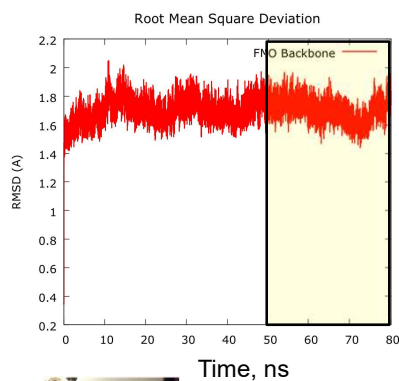
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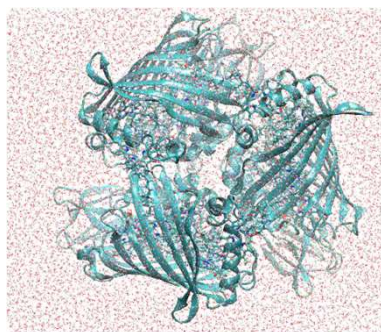
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Example: FMO complex



Yongbin Kim (Slipchenko's group)

- PDB ID: 3ENI
- Classical MD simulations of the FMO trimer: Amber FF, NPT, 80 ns, 300 K, 1 atm
- 100 random snapshots selected from the last 30 ns of MD trajectory

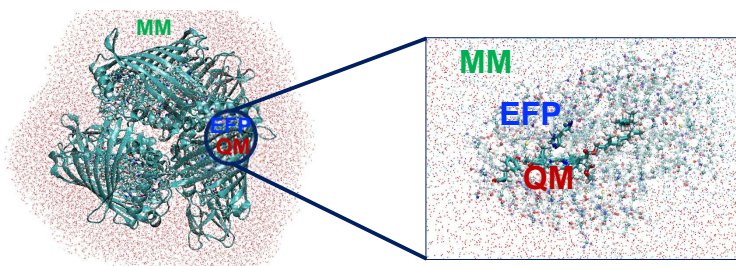


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Example: FMO complex

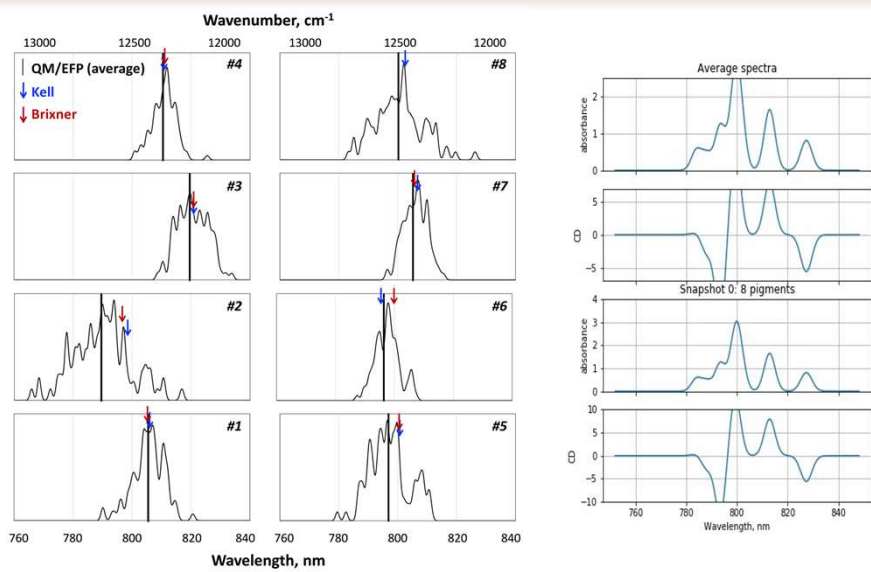
Polarizable embedding simulations: QM/EFP/MM

- QM: each *Bchl a* + Mg coordinating AA, TDDFT-PBE0/6-31G(d)
- **Surrounding:**
within 15 Å of Bchl a head:
polarizable Effective Fragment Potential (EFP)
otherwise:
MM (Amber force field)



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Example: FMO complex



Kell: *J. Phys. Chem. A*, 2016, 120, 6146

Brixner: *Nature*, 2005, 434, 626